

Dae Seok Song

AI Research Engineer | Brain-inspired and Modular Learning

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RESEARCH PROFILE

AI research engineer who has led model development, optimization, clinical-product integration, and international regulatory evidence generation at a medical-AI startup since November 2022. My research asks how a trained neural system can acquire, exchange, and preserve functional modules without global retraining. I combine reproducible experiments with deployment constraints learned from multi-institution medical imaging.

RESEARCH INTERESTS

Brain-inspired learning · Modular and continual learning · Representation alignment · Neural population geometry · Robust learning under distribution shift

INDEPENDENT RESEARCH

RevolverNet: Detachable Functional Modules and Conditional Routing

2026–Present

Independent Researcher

Seoul, Korea

- Designed a modular architecture with interchangeable slots, detachable functional modules, and a router; the project metaphor names these chambers, bullets, and a trigger.
- Across two completed image-classification conditions and 24 directed module transfers, one-time representation alignment recovered 99.4–99.8% of the performance lost under naive swaps. The separately trained models shared the dataset, architecture, and role assignment but used different initializations. Preliminary, unpublished independent work.
- Implemented hard routing so the trigger selects one bullet for each input. Planned work will test whether inputs not covered by existing bullets can indicate when a new chamber and bullet are needed.
- Long-term goal: let people inspect, remove, and replace functional modules while preserving validated behavior and avoiding unnecessary computation.

PROFESSIONAL EXPERIENCE

Connecteve Co., Ltd.

Nov 2022–Present

AI Research Engineer / Early Team Member

Seoul, Korea

- Consolidated nine task-specific models into two uncertainty-aware, partial-label multi-task models, reducing model weights to approximately one-seventh while maintaining target performance across internal evaluations.
- Established the end-to-end MLOps and deployment pipeline; reduced engine size by 50%, accelerated inference by 3×, and limited overall accuracy change to -0.06 percentage points.
- Built a reproducible CT pipeline spanning DICOM anonymization and conversion, PyTorch Lightning training and evaluation, voxel-imbalance handling, and SwinUNETR, DynUNet, and MedDINOv3 experiments.
- Prepared and indexed more than 100,000 medical images for foundation-model studies; evaluated DINOv3, Rad-DINO, EVA-X, linear probing, and parameter-efficient adaptation.

Connecteve Co., Ltd. — Selected Workstreams (continued)

Nov 2022–Present

AI Research Engineer / Early Team Member

Seoul, Korea

- Built a PyTorch Lightning and segmentation-models-pytorch model zoo for pruning, quantization, and architecture comparison; reduced parameters by 1.5–2× at about a one-point performance gap and removed 5.26 GB of stored weights.
- Designed segmentation and classification post-processing, including a random-forest decision layer using softmax and mask features; improved external-site accuracy by approximately 2–3 percentage points.
- Implemented multi-input inference services with TorchServe and FastAPI, including concurrency, error handling, API specifications, and on-premise deployment support.
- Rebuilt the early model architecture and experiment pipeline and conducted 8 classification, 5 detection, and 8 segmentation studies across hospital testbeds, targeting sensitivity and specificity of at least 0.85.
- Led model evidence, technical documentation, and delivery for approvals or regulatory processes in Korea, UAE, EU CE MDR, and Singapore; also supported Japan and U.S. preparation and Deep-Tech TIPS evidence.
- Co-developed the core algorithm and implementation for a peer-reviewed knee-osteoarthritis grading study and its Korean and PCT patent family.

Minerva Soft

Mar 2022–Jul 2022

Image Processing Intern

Korea

- Improved industrial barcode recognition from 70% to 90% and achieved 100–250 ms latency per barcode.
- Built a C++ DLL integration and ran more than 170,000 iterations of memory-leak and stability testing.

EDUCATION

University of Suwon

Mar 2017–Feb 2023

B.S. in Information Security, Valedictorian

GPA 4.29/4.50 · Major 4.37/4.50

Selected coursework: Mathematics I (A+), Algorithms (A+), Data Structures (A0), Computer Architecture (A+), Operating Systems (A+), System Programming (A+).

SELECTED PUBLICATION & RESEARCH OUTPUT

- **Song, D. S.** et al. "Accurate, automated classification of radiographic knee osteoarthritis severity using a novel method of deep learning: Plug-in modules." Knee Surgery & Related Research, 2024. Second author. doi:10.1186/s43019-024-00228-3
- First-author study submitted to KSC 2022: improved segmentation DICE from 86.95 to 90.48 through a task-specific learning pipeline.
- Korean Patent Application 10-2024-0124529 and PCT/KR2025/012689: plug-in modules for automated radiographic knee osteoarthritis severity classification.
- Korean Patent Application 10-2025-0043898: generative knee-image technology.

SELECTED PROJECTS & EARLIER WORK

- **Agricultural price forecasting:** built an LSTM forecasting pipeline (MSE 0.0508; RMSE 0.22539) and presented it in a university AI idea competition.
- **Earlier software projects:** developed Korean conversational-style NLP experiments, an Android stock-chart overlay tool, and a military health-information app selected among the top 50 entries in an open-software competition.

TECHNICAL STRENGTHS

Research	Controlled ablation design, multi-seed evaluation, representation analysis, reproducibility, scientific writing
ML	PyTorch, multi-task learning, uncertainty modeling, medical-image classification/segmentation, domain robustness
Engineering	Python, C/C++, Hydra, model optimization, ONNX/deployment pipelines, MLOps, regulated-product documentation
Experiment systems	Reproducible experiment harnesses, Hydra configuration, automated evaluation, Jira and Confluence documentation

LANGUAGES

Korean (native) · English (working proficiency) · Japanese (conversational)